Status	Finished
Started	Wednesday, 21 October 2026, 3:44 PM
Completed	Wednesday, 21 October 2026, 3:49 PM
Duration	5 mins
Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array

b. When the item is not the array at all

c. When the item is somewhere in the middle of the array ✓

d. Item is the last element in the array or item is not there at all

Question **2**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element

b. The search stops

c. The lower middle element is chosen as the middle element ✗

d. The middle element is chosen randomly

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. There must be a mechanism to access middle element directly

Question **4**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

- a. Sort the list ✗
- b. Compare the target element with the middle element in the list
- c. Compare the target element with the first element in the list
- d. Divide the list into two equal parts

Question **5**

Incorrect

Mark 0.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(n log n) ✗
- b. O🥎
- c. O(1)
- d. O(log n)

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Sorting the list before searching
- c. Dividing the list in half repeatedly
- d. Skipping every second element

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓

b. The list must be sorted

c. There should be direct access to the middle element in any sublist

d. Number values should only be present

Number values should only be present

Question **8**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1

b. n ✓

c. n/2

d. log n

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓

b. O(log n)

c. O(n log n)

d. O(1)

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. At the beginning of the list
c. In the left sublist ✓
d. In the entire list

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Linear search ✓
b. Dynamic search
c. Hashing
d. Binary search

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search is more efficient than binary search on large lists.
c. Linear search requires the list to be sorted.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99

b. 90 and 99 ✓

c. 89 and 94

d. 90 and 94

Question **14**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list

b. Inserting elements into a list

c. Deleting elements from a list

d. Determining whether an element is present in a list ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning

b. The search stops

c. The search continues in the left or right sublist ✓

d. The list is sorted